LGLFormer: Local–Global Lifting Transformer for Remote Sensing Scene Parsing

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***Abstract*— In deep learning, convolutional neural networks (CNNs) and transformers have gained excellent achievements in remote sensing scene parsing. Strong feature representation ability is still a challenge for them. Besides, the complex scenes are still essential challenges for deep learning in remote sensing scene parsing. In this article, an efficient local–global lifting transformer (LGLFormer) framework is proposed to ease the challenges above. It effectively combines CNNs, transformer, and wavelet transform to build a strong local–global (LG) feature representation network. Besides, global feature learning driven by LG adaptive features is proposed based on the 2-D LG adaptive feature extractor (LGAFE) and refined global feature attention module. The 2-D LG lifting feature extractor is inspired by the lifting scheme, which introduces local and global dependency. Furthermore, two LG lifting schemes are proposed, including the series and parallel modes, which can effectively learn LG relations between pixels. Finally, experiments are validated on three remote sensing benchmark datasets. The proposed LGLFormer achieves the state-of-the-art with 99.02%, 99.2%, and 99.48% overall accuracy (OA) on AID, WHU-RS19, and UCM datasets, respectively. In addition, LGLFormer shows good convergence with competitive parameters. The experimental code will be available at https://github.com/yutinyang/LGLFormer.**

***Index Terms*— Convolutional neural network (CNN), lifting scheme, local–global (LG), remote sensing, transformer, wavelet.**

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I. Introduction

CONVOLUTIONAL neural networks (CNNs) and transformers occupy essential positions in deep learning [1]. They are two crucial feature extractors that are used for deep visual representation [2]. The former can realize the modeling between local adjacent pixels but lacks global modeling capabilities. The latter owns strong global modeling ability and is developed rapidly in deep learning tasks [3]. Transformer first achieves excellent results in natural language processing (NLP) [4]. The appearance of ViT [5] is a preliminary attempt by the transformer in image processing and achieved performance comparable to CNNs. Subsequently, a large number of transformer models, such as BotNet [6], Swin transformer [7], and CSwin transformer [8], showed superiority in image classification, target detection, image segmentation, and other fields. Researchers try to combine CNNs and transformers to integrate local and global representation capabilities to achieve more robust representation, including [9], [10], and so on. Fang et al. [9] presented a hybrid framework for image super-resolution, and Luo et al. [10] proposed the integration of CNN and transformer for medical image segmentation. Of course, there are some attempts at remote sensing scene parsing, such as the efficient multiscale transformer and cross-level attention learning (EMTCAL) [11], the explainable spatial–frequency multiscale transformer (SF-MSFormer) [12], and so on. However, realizing more robust feature representation still owns important research significance.

Although existing methods pay significant attention to remote sensing scene data features, robust feature representation networks are still required to handle complex scenes. For instance, the example of a complex remote sensing scene is depicted in Fig. 1, which comprises various objects such as bridges, buildings, trees, and roads. Moreover, these objects are situated at different distances from one another. When observing other areas, additional targets are excluded, which plays a crucial role in the final scene recognition. Specifically, the river in the local area aids in bridge scene recognition, while roads, trees, and buildings on land fall into the global visual area. In addition, the local and global relationships between different pixels carry different significance for scene recognition.

Wavelet transform can capture time–frequency, multiscale, and multiresolution features [13]. It is considered to provide features from coarse to refined and offers refined features

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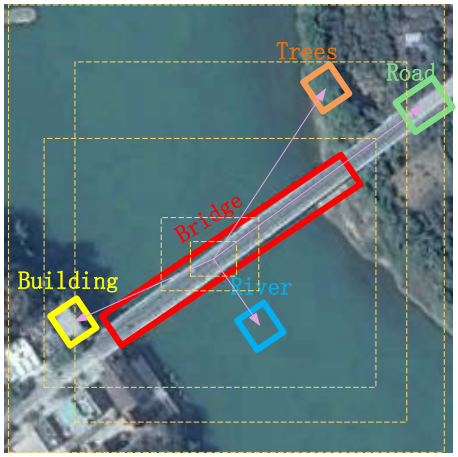


Fig. 1. Example of complex remote sensing scene. The dotted lined boxes in yellow represent the visual receptive fields in different sizes. The pink lines with arrows represent the distance of different objects to the bridge.

with multisubbands [14]. Several wavelet-related works integrate wavelet transform with CNNs to achieve better feature representation. For example, wavelet CNNs are built based on multiresolution analysis and CNNs, which directly utilize the wavelet transform to represent the input for CNNs learning [15]. Dual wavelet attention networks (DWANs) rethink the traditional channel and spatial attention by incorporating wavelet transform [16]. This approach aims to address the problem of information loss in conventional attention mechanisms and enhance the feature representation of CNNs [17]. However, using a fixed wavelet is likely suboptimal as it restricts adaptability and cannot leverage data-driven learning.

The lifting scheme makes the wavelet transform not limited to the fixed wavelet. Deep adaptive wavelet networks, DAWN, proposes using the adaptive lifting scheme to construct the multiresolution analysis networks [18]. In detail, it mainly uses convolution to improve the lifting scheme to design deep adaptive wavelet networks. In addition, Yang et al. [12] proposed SF_MSFormer that combining the adaptive lifting scheme and self-attention can capture the texture-enhanced features in remote sensing scenes.

However, both DAWN [18] and SF_MSFormer [12] mainly use convolution operations to design the updater and predictor in the lifting scheme. Convolution operations especially obtain local receptive fields and lack global dependence. Inspired by this, the global dependency is considered in the lifting scheme to achieve better and refined feature representation. The 2-D adaptive local–global (LG) feature extractor is proposed to obtain the LG relation.

Based on the above, the LG lifting transformer (LGLFormer) is proposed, which combines CNNs, a transformer, and a 2-D adaptive LG feature extractor for remote sensing scene parsing. Two types of 2-D adaptive LG feature extractors are introduced, including the series and parallel modes. Experimental results demonstrate that the proposed LGLFormer achieves state-of-the-art performance on three benchmark datasets. In summary, the contributions of this article are concluded as follows.

1) The LGLFormer network is proposed. It achieves strong LG feature representation learning for remote sensing scene parsing.
2) The global dependence is considered in the lifting scheme, which can promote obtaining finer features.
3) Two adaptive and effective lifting modules are proposed, including the global and series/parallel LG lifting schemes. They can effectively enhance the feature representation of CNNs.
4) The proposed method achieves state-of-the-art performance with significantly fewer parameters and complexity, leading to improved efficiency in remote sensing applications.

The remainder of this article is organized as follows. Section II presents the related work. Section III presents the LGLFormer for scene parsing. Section IV displays the experimental results and gives the analysis about the proposed methods. Finally, Section V concludes this article and gives further studying direction.

II. Related Work

In this section, the related work on wavelet transform, CNNs, and transformer representation are introduced.

A. Wavelet Transform and Lifting Scheme

The lifting scheme is also known as the second-generation wavelet. It inherits the advantages and also increases the speed of wavelet decomposition. Its framework makes itself adaptive by designing the predictor and updater, which makes it flexible but does not depend on a specific wavelet basis [19].

For an input signal x, the lifting scheme can generate the approximation coefficients L and the details coefficients H. The lifting scheme mainly contains three stages: splitting, updating, and predicting. First, the splitting operation initially splits the input into two nonoverlapping partitions. The easiest splitting method is to divide the signal according to the odd lines (columns) and the even lines (columns), known as the lazy wavelet division method. The updater primarily performs frequency domain separation, where the odd columns are combined with the updating value obtained from the even columns to form the approximate average L. The basic idea of the predictor is that the two divided parts are closely related, allowing one part to predict the other by tracking the differences/details H between them.

Zheng et al. [20] proposed using nonlinear neural network functions to design the updaters and predictors. Besides, the training loss of the predictor and updater was considered during training. Rodriguez et al. [18] expand the idea of Zheng et al. [20] into the classification task. They propose DAWN, which uses the 2-D adaptive lifting scheme to build a multiresolution neural network. It realizes the end-to-end wavelet convolutional network training. Later, Yang et al. [12] also used the 2-D adaptive lifting scheme and CNNs to construct SF_MSFormer. It shows that using CNNs to develop updaters and predictors effectively obtains the adaptive features.

Besides, the wavelet convolutional networks have been well developed. Many wavelet-related convolution networks are appearing, such as DAWN [18], DWAN [16],

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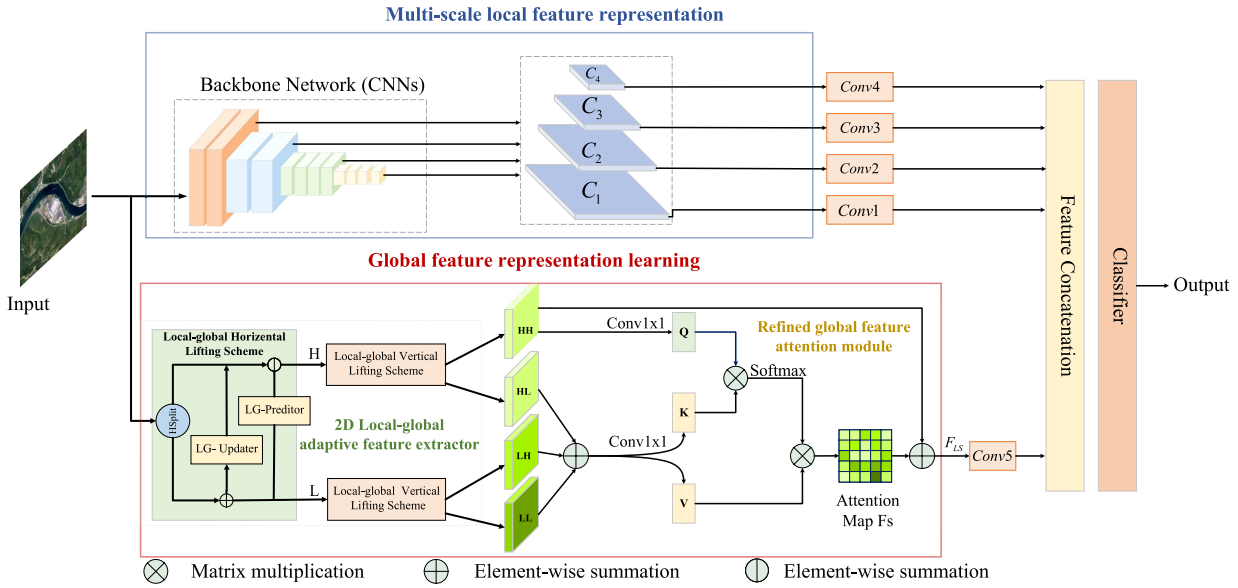


Fig. 2. Overall architecture of the L^{2}GFormer for remote sensing scene parsing. HSplit represents the horizontal splitting method. VSplit represents the vertical splitting method.

SF_MSFormer [12], Wavelet CNN [15], and so on. In addition, it has been considered for different deep learning tasks, including image classification [18], [21], image super-resolution [22], compression [23], and so on.

B. CNNs and Transformer

CNNs have achieved excellent results in deep learning due to their strong local feature extraction capabilities [24]. CNNs, recurrent neural networks (RNNs), and transformers are three commonly used feature extractors. Among them, CNNs and transformers have made rapid advancements in visual representation and have gained significant prominence.

CNNs are network structures based on convolutional and pooling layers primarily used in image processing. The convolutional layer can extract spatial features from images, such as edges and corner points. The pooling layer, which follows the convolutional layer, further reduces the size of feature maps and decreases computational complexity. CNNs are capable of classifying and recognizing complex images through multilayer convolutional and fully connected layers.

Transformer is a self-attention-based feature extraction network initially developed in NLP. The self-attention mechanism can calculate each position in the input sequence to obtain global context information [25]. Transformer's encoder–decoder structure can implement machine translation and text generation tasks. Subsequently, the transformer was extended to visual representation because of its strong global modeling ability [26]. The ViT tried a transformer for visual recognition for the first time and obtained a competitive performance compared with CNNs [5].

CNNs primarily focus on local modeling, as convolutional operations only consider the relationship between neighboring pixels [27]. In contrast, transformers prioritize global processing. Their self-attention mechanism eliminates the need for the local hypothesis assumed by CNNs. Transformers require more data during training. Therefore, combining CNNs and transformers can provide mutual benefits. This combined model has become a research direction for deep learning enthusiasts. Researchers have transferred successful experiences from CNNs to transformers, improving their performance and efficiency. For instance, several enhanced models have been proposed, including direct replacement of position encoding (CPVT [28] and ResT [29]), structural combinations (CoAtNet [30]), local processing (CeiT [31]), and others.

Furthermore, CNNs can be directly combined with transformer to build a neural network model architecture by leveraging the strengths of these two frameworks. Similar to previous attention-based approaches, some researchers have attempted to integrate transformer into CNNs by replacing certain convolutional blocks with transformer layers, such as CvT [32] and BotNet [6]. In summary, integrating CNNs and transformers remains highly discussed and researched.

III. LGLFormer for Remote Sensing Scene Parsing

The LGLFormer is proposed for remote sensing scene parsing. The overall architecture of it is introduced at first. Then, the proposed 2-D LG adaptive feature extractor (LGAFE) is detailed based on the LG lifting scheme.

A. Overall Architecture of LGLFormer

The overall architecture of LGLFormer is shown in Fig. 2. It consists of two branches: the multiscale local feature representation branch and the global feature learning branch. CNNs are used to extract the multiscale local feature representation. Simultaneously, the input is forwarded to the second branch for global feature learning, incorporating LG adaptive features. Finally, the output features from these two branches are concatenated for final recognition. The entire process can be formulated as follows.

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For an input x, the multiscale local feature representation branch can be expressed as

$$C_i = \mathrm{MS}(x) \tag{1}$$

where MS represents the multiscale feature representation of the backbone network, and C_i represent the feature maps at the ith scale ($i = 1, 2, 3, 4$).

For the global feature representation learning branch, the 2-D LGAFE can obtain the features HH, HL, LH, and LL

$$[\mathrm{HH},\mathrm{HL},\mathrm{LH},\mathrm{LL}] = \mathrm{LGAFE}(x). \tag{2}$$

Then, the outputs of the 2-D LGAFE are sent to the refined global feature attention module. It is based on a transformer to calculate the attention maps F_s. In detail, the query (Q), key (K), and value (V) are set as

$$Q = \mathrm{LL};\ K = V = \mathrm{LL} + \mathrm{LH} + \mathrm{HL} + \mathrm{HH}. \tag{3}$$

Due to the proximity of most detail values to zero, the combination of low-frequency and high-frequency subbands can either enhance or weaken the context features [33]. By setting $Q = \mathrm{LL}$ and $K = V = \mathrm{LL} + \mathrm{LH} + \mathrm{HL} + \mathrm{HH}$ as inputs for self-attention, a texture-enhanced feature representation can be achieved [12]. In detail, the attention feature maps F_s can be expressed as

$$F_s = \mathrm{Softmax}\left(\left(\mathrm{QK}^T\right)/\sqrt{d}\right)V \tag{4}$$

where $1/(d)^{1/2}$ represents the scaling factor in self-attention.

The output of the refined feature attention module keeps the skip connection as $F_{\mathrm{LS}} = \mathrm{LL} + F_s$. For the final recognition, the multiscale local feature representation branch's output $C1$, $C2$, $C3$, $C4$, and the output of the global feature representation learning branch' output F_{LS} are concatenated together. In detail, Conv1($C1$), Conv2($C2$), Conv3($C3$), Conv4($C4$), and Conv5(F_{LS}) are concatenated together and then sent to the classifier to output the final category.

B. Two-Dimensional LGAFE

The proposed 2-D LGAFE is inspired by the 2-D adaptive lifting scheme. It mainly proposes to design the updator and predictor with convolution and self-attention, which can introduce local and global representation learning ability in the lifting scheme. As shown in Fig. 3, the process of the 2-D LGAFE can be formulated as follows.

For an input x, the 2-D LG adaptive lifting scheme consists of one LG horizontal lifting scheme following two LG vertical lifting scheme modules. First, the input is split into two parts as

$$[x1, x2] = H\mathrm{Split}(x) \tag{5}$$

where $x1$ and $x2$ are two nonoverlapping partitions and HSplit represents the horizontal splitting. In the LG horizontal lifting scheme, the horizontal splitting method is adapted. In detail, the odd row and even row are the splitting results. The approximation L and the detail H are expressed as

$$\begin{aligned} L &= x1 + \text{LG-Updater}(x2) \\ H &= x2 - \text{LG-Predictor}(x1) \end{aligned} \tag{6}$$

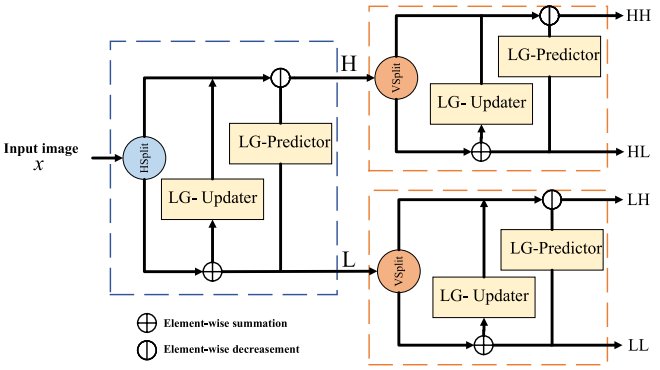


Fig. 3. Two-dimensional LGAFE. LG-Updater and LG-predictor represent the LG updater and predictor, respectively. HSplit/VSplit represents the horizontal/vertical splitting method.

where LG-Updater and LG-Predictor represent the LG updater and the LG predictor, respectively. Then, the L and H are sent to the LG vertical lifting scheme module. The LG vertical lifting scheme module splits them into two parts too, which can be formulated as

$$\begin{aligned} [\mathrm{LL0}, \mathrm{LH0}] &= V\mathrm{Split}(L) \\ [\mathrm{HL0}, \mathrm{HH0}] &= V\mathrm{Split}(H) \end{aligned} \tag{7}$$

where VSplit represents the vertical splitting. In the LG horizontal lifting scheme, the odd column and even column are the splitting results. Then, the outputs of the LG horizontal lifting scheme can be expressed as

$$\begin{aligned} \mathrm{LL} &= \mathrm{LL0} + \text{LG-Updater}(\mathrm{LH0}) \\ \mathrm{LH} &= \mathrm{LH0} - \text{LG-Predictor}(\mathrm{LL0}) \\ \mathrm{HL} &= \mathrm{HL0} + \text{LG-Updater}(\mathrm{HH0}) \\ \mathrm{HH} &= \mathrm{HH0} - \text{LG-Predictor}(\mathrm{HL0}) \end{aligned} \tag{8}$$

where LL represents the low-frequency component and LH, HL, and HH represent three high-frequency subbands.

C. Global/LG Lifting Scheme

The global modeling is considered in the 2-D adaptive feature extractor. The LG lifting scheme is proposed here. There are mainly three modes for the updater and predictor, including the global lifting updater/predictor, the LG lifting updater/predictor in series or parallel connection mode, as shown in Figs. 4–6.

1) Global Updator/Predictor: To introduce the global independency to the lifting scheme, the self-attention is used to capture the global features during the updating and predicting processes (as shown in Fig. 4).

For the input x, the global lifting module is formulated as follows:

$$q = w_3x;\ k = w_4x;\ v = w_5x \tag{9}$$

where w_3, w_4, and w_5 represent the convolution operations of Conv_3, Conv_4, Con_5 in Fig. 4, respectively,

$$P = U = x + \mathrm{softmax}\left(\left(\mathrm{qk}^T\right)/\sqrt{d}\right)v. \tag{10}$$

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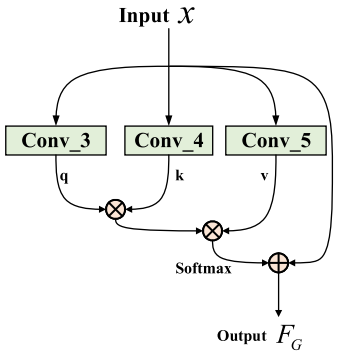


Fig. 4. Global lifting updater/predictor.

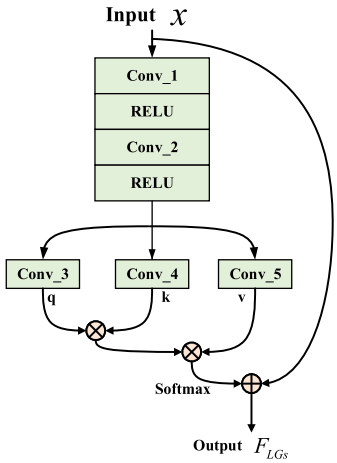


Fig. 5. Series LG lifting updater/predictor.

The proposed global lifting scheme can retain global independence during the lifting.

2) LG Updator/Predictor: There are two lifting schemes that incorporate local and global modeling during the lifting process, allowing for the retention of local and global feature relationships. In detail, the two modules are shown in Figs. 5 and 6. The convolution operations and self-attention are utilized in both series and parallel connection modes.

As shown in Fig. 5, the LG updator/predictor in series can be formulated as

$$g(x) = \delta(w_2(\delta(w_1 x))) \tag{11}$$

where δ represents the RELU activation operation. And then

$$q' = w_3 g(x); k' = w_4 g(x); v' = w_5 g(x). \tag{12}$$

Finally, the series connection mode's output can be formulated as

$$P(x) = U(x) = x + \text{softmax}\left(\left(\text{q}'\text{k}'^T\right)/\sqrt{d}\right)v'. \tag{13}$$

For the LG lifting scheme in parallel, the updator/predictor can be figured in Fig. 6. In detail, it can be formulated as

$$P(x) = U(x) = \text{softmax}\left(\left(\text{qk}^T\right)/\sqrt{d}\right)v * g(x) + x. \tag{14}$$

When the model prioritizes local features, it focuses on the local details and patterns of the input data, capturing features

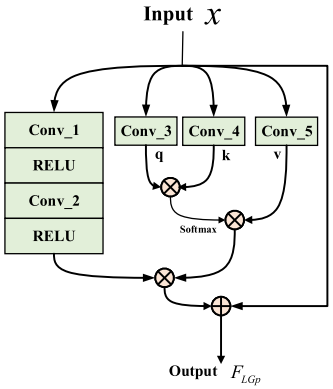


Fig. 6. Parallel LG lifting updater/predictor.

such as local textures, shapes, and structures. Subsequently, by propagating and integrating these local features, the model gradually gains an understanding of the global context. The approach of simultaneously considering global-local features combines these two methods. It aims to incorporate both global and local features during processing to capture their relationships and interactions more effectively. This approach provides richer contextual information and helps the model develop a more comprehensive understanding of the data.

D. Design Details

As for the loss, the proposed framework utilizes the standard cross-entropy loss during training.

When implementing the proposed methods, the convolution operations and architecture details are introduced here. All of the Conv1, Conv2, Conv3, Conv4, and Conv5 in the framework are adaptive average pooling operations, ensuring that the output of different branches is of the same size (1×1) for feature concatenation.

For the local lifting scheme, a convolution operation with a kernel size of 4×4 is employed to capture the local relationships between pixels. In the global lifting scheme, a convolution kernel size of 1×1 is used to consider global dependencies. In the series mode of the LG lifting scheme, a self-attention operation is inserted after the local convolution lifting process. In the parallel mode of the LG lifting scheme, the self-attention operation is directly applied to the splitting signal without additional convolution operations, in addition to the parallel local lifting scheme branch. Specifically, the kernel size in the three modes of Conv_1 is set to 4×4, while the kernel size for Conv_2, Conv_3, Conv_4, and Conv_5 is set to 1×1.

IV. Experimental Results and Analysis

The experimental datasets and settings are detailed in this section. In addition, experimental results about the model's performance are shown and analyzed to determine the effectiveness of the proposed method.

A. Experimental Settings

1) Datasets: The AID dataset [34], UCM dataset [48], and WHU-RS19 dataset [49] are selected as benchmarks to

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Fig. 7. Example samples of experimental datasets, including (a) AID, (b) UCM, and (c) WHU-RS19 datasets.

validate LGLFormer. The detailed information on these three datasets is as follows.

The images in the AID dataset [34], UCM dataset [48], and WHU-RS19 dataset [49] are collected from Google Earth, Google Satellite, and the USGU National Map Urban Area Image Collection, respectively. The AID dataset consists of 10 000 images with 30 classes, the UCM dataset consists of 2100 images with 21 classes, and the WHU-RS19 dataset consists of 1005 images with 19 classes of remote-sensing scenes. The image size is 600×600 for AID [34] and WHU-RS19 [49], while for UCM [48], the image size is 256×256.

Moreover, the spatial resolution of the AID dataset [34], UCM dataset [48], and WHU-RS19 dataset [49] is approximately 0.5, 0.8, 0.3 m, and up to 0.5 m, respectively. Examples and detailed categories of scenes from these benchmark datasets are shown in Fig. 7.

During data processing, random cropping, horizontal flipping, and normalization operations are applied. All images in the three datasets are resized to 224×224 for training and testing. The images in the UCM dataset are directly fed into the proposed network with their original size of 256×256.

2) Settings: The stochastic gradient descent (SGD) optimizer with a momentum of 0.9 is used for training. All the experiments are trained for 300 epochs. The learning rate is set at 0.01 with the learning rate decay at the 150 and 200 epochs. All the models are trained on the NVIDIA Titan V100 4 GPUs with the framework of Pytorch 1.7.1.

3) Evaluation Metrics: The proposed LGLFormer mainly experiments with the remote sensing scene classification. The OA and the confusion matrix are metrics to evaluate the model's performance. OA is defined as the ratio of correctly classified images divided by the total number of testing images. The confusion matrix can display the classification accuracy of each scene category, showing whether the prediction is correct or mistaken for other scene categories. Besides, the models' computation complexity is assessed with the parameters and GFlops indicators.

4) Contrasting Methods: The proposed LGLFormer is a multiscale method that incorporates LG representations. It combines CNNs and the transformer's self-attention within its framework. Thus, several contrasting methods are considered, including models that focus on multiscale, LG representations, CNNs, and transformer-related techniques. In addition, the LG representation in LGLFormer is primarily based on

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TABLE I

COMPARE OVERALL ACCURACY (OA) (%) ON THE BENCHMARK DATASETS (50% FOR TRAINING) OF THE CONTRASTING METHODS. THE BEST PERFORMANCE IS IN BOLD

Datasets / Methods	Publication (Years)	AID Ave.	AID Std.	WHU-RS19 Ave.	WHU-RS19 Std.	UCM Ave.	UCM Std.	Parameters Unit(M)	GFlops Unit(G)
CaffeNet [34]	TGRS (2017)	89.53	0.31	96.24	0.67	93.98	0.67	-	-
GoogleNet [34]	TGRS (2017)	86.39	0.55	94.71	0.75	92.70	0.6	7	0.75
VGG-VD-16 [34]	TGRS (2017)	89.64	0.36	96.05	0.42	94.14	0.69	138.36	7.75
ResNet18 [35]	TNNLS (2022)	95.51	0.31	-	-	97.43	0.19	11.19	1.84
CNN-AE [36]	IJRS (2016)	92.03	1.36	92.58	1.07	92.06	1.57	-	-
CNN-NN [36]	IJRS (2016)	92.97	1.45	93.31	1.24	93.49	2.54	-	-
DCA-Fusion [37]	TGRS (2017)	94.33	1.52	93.56	0.96	93.84	0.79	-	-
GM [38]	TGRS (2018)	89.22	1.86	88.15	0.88	88.77	1.55	-	-
GLM512 [38]	TGRS (2018)	91.98	1.34	90.23	0.79	90.12	1.27	-	-
GLM16 [38]	TGRS (2018)	93.45	1.52	92.99	0.83	92.60	1.31	-	-
ResNet18$_{global+local}$ [35]	TNNLS (2022)	96.76	0.20	-	-	97.95	0.05	-	-
Yuan et.al [38]	TGRS (2018)	95.29	1.28	95.76	0.74	94.97	1.16	0.6	-
APDC-Net [39]	GRSL (2019)	92.15	0.29	-	-	95.01	0.43	-	-
RSFJR[40]	TGRS(2019)	96.18	1.36	97.48	0.85	97.21	0.65	550.98	-
Wavelet CNN[41]	TNNLS(2020)	96.65	0.24	86.2	0.68	92.19	0.47	10.24	2.98
C-CNN[41]	TNNLS(2020)	97.36	0.45	-		-	-	-	-
DWAN [16]	TCSVT (2022)	94.50	0.3	-	-	96.19	0.13	-	-
LCNN-BFF [42]	J-STARS (2020)	94.62	0.16	-	-	94.64	0.21	6	-
MSA-Network [43]	J-STARS (2021)	96.01	0.43	-	-	97.8	0.33		
EFPN-DSE-TDFF [44]	TGRS (2021)	94.50	0.3	-	-	96.19	0.13	-	-
MS2AP[45]	Neurocomputing(2021)	96.46	0.20	98.88	0.40	-	-	23.7	-
MF^{2}CNet[46]	TGRS(2022)	97.02	0.28	-	-	98.76	0.18	30.12	2.42
ViT [11]	TGRS (2022)	96.49	0.18	-	-	98.48	0.39	49	10.1
SCViT [47]	TGRS (2022)	96.98	0.16	-	-	98.9	0.19	40.1	9.3
EMTCAL* [11]	TGRS (2022)	97.49	0.36	97.6	0.23	98.67	0.16	27.0	4.23
SF-MSFormer[12]	TGRS (2023)	98.57	0.23	97.8	0.39	99.35	0.26	26.12	19.56
LGLFormer(Ours)	(2023)	**99.02**	0.11	**99.2**	0.06	**99.48**	0.04	22.96	14.29

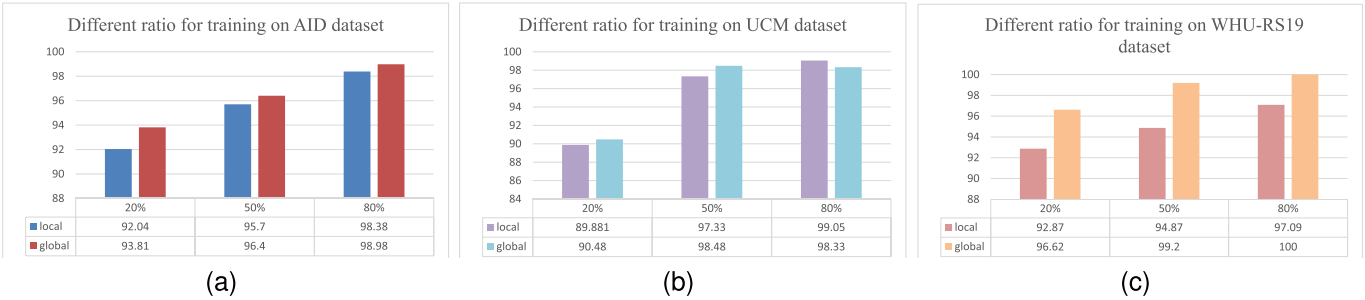


Fig. 8. Compare the accuracy of local and global lifting schemes in the proposed methods. (a) Compare the accuracy of the local and global lifting scheme on the AID dataset. (b) Compare the accuracy of the local and global lifting scheme on the UCM dataset. (c) Compare the accuracy of the local and global lifting scheme on the WHU-RS19 dataset.

the lifting scheme. Consequently, the performance of various wavelet-related methods is shown here.

In detail, the multiscale-related methods include the EFPN-DSE-TDFF [44], the multiscale stacking attention pooling (MS2AP) [45], the multiscale feature fusion covariance network (MF^{2}CNet) [46], and the multiscale network (MSA-Network) [43]. The LG representation contrasting methods mention GM [38], GLM16 [38], and GLM512 [38]. GM [38] method combines the global feature with multilayer perceptron. GLM16 [38] and GLM512 [38] introduced 16 and 512 local features to the global feature representation for final representation. The convolution neural networks includes CaffeNet [50], GoogleNet [50], VGG-VD-16 [51], CNNs-AE [36], and CNNs-NN [36]. Some wavelet-based methods are selected as the contrasting methods, including Wavelet CNN [41], C-CNN [41], DWAN [16], and so on. ViT [50] and SCViT [50], EMTCAL [11], and SF_MSFormer [12] are some transformer-related models for contrasting.

B. Experimental Results

L^{2}GFormer is mainly experimented on the WHU-RS19, UCM, and AID datasets with 50% for training, compared with the related contrasting methods. Their performances are shown in Table I.

1) Performance on WHU-RS19: Transformer-based methods (ViT [50], SCViT [50], and EMTCAL [11]) show increased accuracy than most of the CNN-related methods. In detail, the RSFJR [40] obtains about 97.48% OA. The performances on WHU-RS19 of the contrasting methods show that the MS2AP [45] shows 98.88% OA with 23.7M parameters. It offers better performance than the other contrasting

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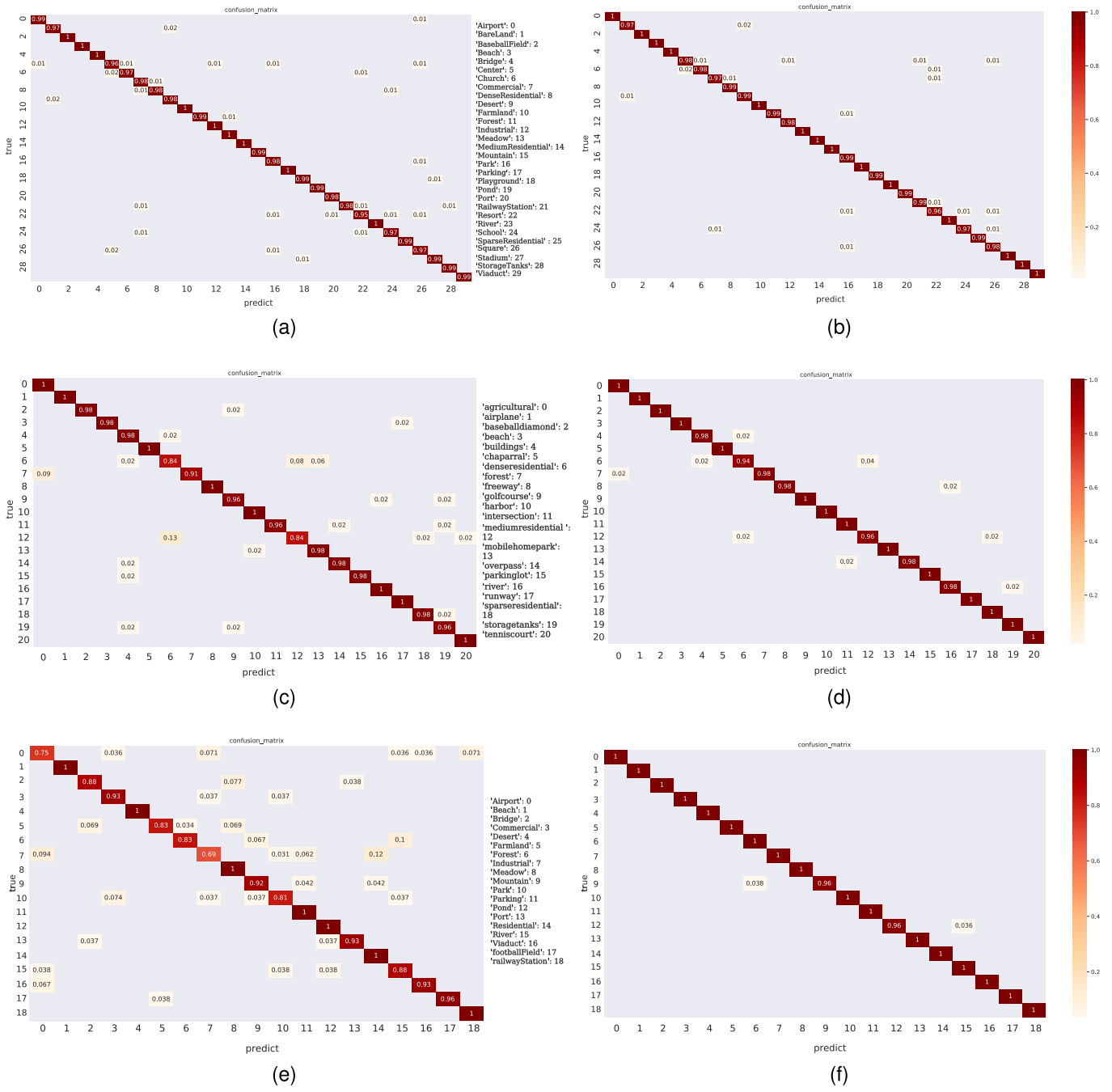


Fig. 9. Confusion matrices of LGLFormer on the three benchmark datasets (50% for testing). (a) LGLFormer with local lifting scheme on AID. (b) LGLFormer with LG lifting scheme on AID dataset. (c) LGLFormer with local lifting scheme on UCM. (d) LGLFormer LG lifting scheme on UCM. (e) LGLFormer with local lifting scheme on WHU-RS19. (f) LGLFormer LG lifting scheme on WHU-RS19.

methods. The proposed LGLFormer with series mode shows about 99.2% OA with only 22.96M parameters. By contrast, LGLFormer with series mode shows about 0.32% increased OA without increasing parameters.

2) Performance on UCM: As shown, ResNet18 [35] shows about 97.43% OA on the UCM dataset. It performs better than the traditional CNNs models, such as the CaffeNet [34], GoogleNet [34], and VGG-VD-16 [34]. Global methods with 16 and 512 local features perform better than those without local features. Besides, Global methods with 16 local features show about 2.76% increased accuracy than that with 512 local features. It shows that the number of local features is not proportional to the model performance. In addition, ResNet18 with LG features shows about 0.52% increased OA than ResNet18. Besides, ResNet18 with LG features performs better than the LG representation methods GLM16 [38].

The multiscale and transformer-related methods (EMTCAL* [11] and SF-MSFormer [12]) proposed in 2023 achieve excellent performances on the UCM dataset. By contrast, the proposed LGLFormer achieves the state-of-the-art with 0.81% and 0.13% increased accuracy with 4.04M and 3.16M decreased parameters.

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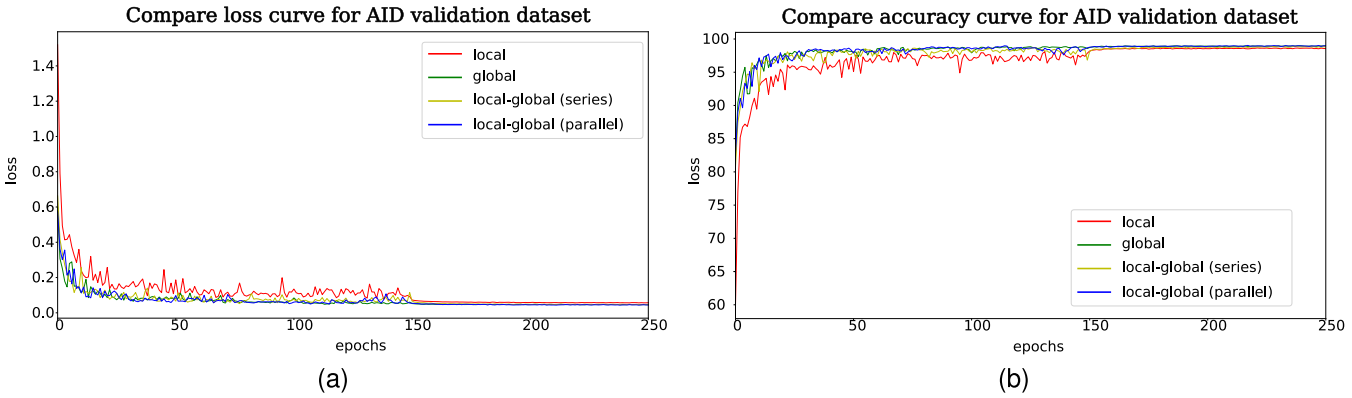


Fig. 10. Compare accuracy and loss curves on the AID validation dataset. (a) Compare loss curves on the AID validation dataset. (b) Compare accuracy curves on the AID validation dataset. The local LGLFormer curves in red, global in green, LG (series) in yellow, and LG (parallel) in blue.

3) Performance on AID: The compare accuracy of the proposed LGLFormer and the contrasting methods on the AID dataset are shown in Table I. It shows that LGLFormer still achieves the state-of-the-art on the AID datasets. Compared with CaffeNet [34], GoogleNet [34], and VGG-VD-16 [34], ResNet18 shows about 5.87%–9.12% increased accuracy. Global methods show competitive performance with CaffeNet [34] and VGG-VD-16 [34]. Global methods with 16 local features show 4.23% increased OA than Global methods. In addition, the Wavelet CNN [41], C-CNN [41] shows better precision than DWAN [16].

ViT [11], SCViT [47], EMTCAL* [11], and SF-MSFormer [12] shows successful performance on AID dataset. Among them, SF-MSFormer [12] performs better because of effectively combining the CNNs and transformer with multiscale representation and texture-enhanced consideration. The proposed LGLFormer obtains excellent performance with 99.02% OA on the AID dataset.

TABLE II

COMPARE OA (%) OF THREE MODES OF THE LIFTING SCHEMES (LFT) WITH RESNET18 BACKBONE WHEN 20%, 50%, AND 80% RATIO FOR TRAINING DATASETS

DataSets	AID			WHU-RS19			UCM		
Methods	20%	50%	80%	20%	50%	80%	20%	50%	80%
onlyglobal	93.81	96.4	98.98	96.62	99.2	**100**	90.48	98.48	98.33
series	93.8	99.02	98.92	**96.75**	99.2	**100**	**92.26**	**99.48**	99.05
parallel	**94.06**	**99.04**	**99.05**	96.63	**99.6**	99.52	90.12	99.05	**99.81**

TABLE III

COMPARE OA (%) OF LGLFormer WITH DIFFERENT BACKBONES

Backbones	AID	WHU-RS19	UCM
FPN(ResNet18)	98.38	94.87	97.33
ResNet18	99.02	99.2	**99.48**
ResNet34	**99.19**	**99.6**	98.81

C. Ablations

1) Local Versus Global Lifting Scheme: The performances of the proposed framework with only local or global lifting schemes are shown. As shown in Fig. 8, the framework with the global lifting scheme performs better than that with the previous convolution-based local lifting scheme. As Fig. 8 shows, it is clear that the global lifting scheme shows nearly increased OA on all of the benchmark datasets, including AID, UCM, and WHU-RS19. In detail, LGLFormer with global lifting scheme shows about 96.4%, 98.48%, and 99.2% OA when 50% of the datasets are used for training. However, LGLFormer with local lifting scheme only achieves 95.7%, 97.33%, and 94.87% OA on the AID, UCM, and WHU-RS19 datasets, respectively. Experimental results with 20%, 50%, and 80% training data show increased accuracy with the global lifting scheme mode except for when 80% of UCM for training.

2) Series Versus Parallel LG Lifting Scheme: Two LG lifting schemes are proposed in this article, including the series LG lifting scheme and the parallel mode. The performances of these two modes are shown in Table II. It shows that the series and the parallel LG lifting scheme perform better than the global one. However, the parallel LG lifting scheme achieves state-of-the-art performance on the AID dataset with 94.06%, 99.04%, and 99.05% OA on 20%, 50%, and 80% training data, respectively. On the other hand, the series LG lifting scheme outperforms the parallel lifting scheme on the WHU-RS and UCM datasets.

The series and parallel LG lifting schemes have varying effects on different types of data and tasks. In some cases, prioritizing local features may be more suitable for highlighting local details and structures, while prioritizing global features may be more appropriate for emphasizing the overall context and semantics. Simultaneously considering both global and local features can strike a balance between these aspects, but its effectiveness may vary depending on the characteristics of the data and the requirements of the task.

3) Ablation on the Backbone Network: To evaluate the multiscale features branch, the backbone network is ablated here. Feature pyramid networks (FPNs), ResNet18, and ResNet34 are selected for the ablation. As shown in Table III, the framework with only the ResNet18 backbone offers an increase in accuracy of about 0.64%, 4.33%, and 2.15% on the AID, WHU-RS19, and UCM datasets, respectively. The ResNet18 backbone demonstrates better performance compared to FPN

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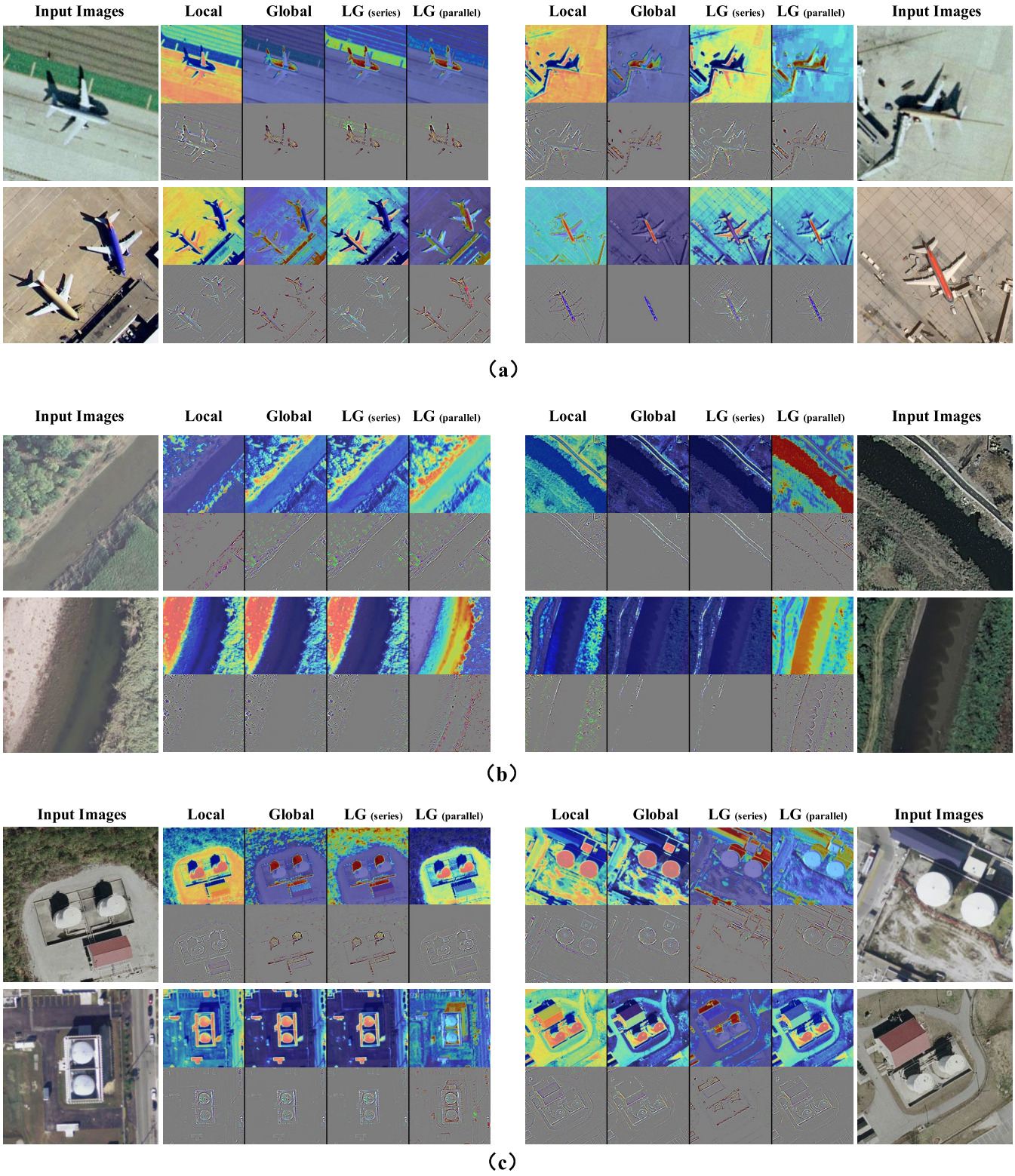


Fig. 11. Feature visualizations of the proposed method with local lifting scheme, global lifting scheme, and the LG lifting scheme in series or parallel. (a) Feature visualization of airplanes. (b) Feature visualization of rivers. (c) Feature visualization of storage tanks. There are two feature visualization methods are used, including the CAM-class activation mapping (top) and the GradCAM-gradient-weighted class activation mapping (behind).

and achieves competitive performance compared to ResNet34. These results suggest that ResNet18 provides sufficient multiscale representation features for the proposed framework. In addition, it is evident that selecting ResNet18 as the backbone strikes a better balance between accuracy and the number of parameters.

4) Ablation on the Global Feature Representation Learning Branch: ViT has been a common method for global representation and has once again demonstrated good performance in image classification, including remote sensing. Therefore, ViT was selected as the global representation branch for comparison with the proposed global feature representation

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TABLE IV
COMPARE ACCURACY OF THE GLOBAL FEATURE BRANCH AND VIT

Data Sets	AID		WHU-RS19		UCM	
Methods	OA(%)	Para.(M)	OA(%)	Para.(M)	OA(%)	Para.(M)
with ViT	98.73	76.249	97.8	76.235	97.24	76.237
Ours	**99.02**	22.575	**99.2**	22.563	**99.48**	22.565

TABLE V
COMPARE PARAMETERS AND GFLOPS OF THE PROPOSED METHODS

Methods	Parameters(M)	GFlops
local	22.534	11.479
global	22.657	14.048
local-global (series)	22.956	14.292
local-global(parallel)	**22.575**	**11.993**

learning branch. The compare overall accuracies (%) are shown in Table IV.

In detail, the comparison includes the OA and training parameters (para.) of the proposed method and the one with the ViT branch replacing the global feature representation branch. The results indicate that the proposed method achieves better performance with fewer parameters compared to the ViT branch. The global feature representation consists of a 2-D LGAFE and a simple refined global feature attention module. The comparison of experimental results further supports the effectiveness and superiority of the proposed lifting scheme.

D. Confusion Matrix and Convergence Analysis

The confusion matrices of LGLFormer are evaluated on three datasets (50% for testing), shown in Fig. 9. The confusion matrices of LGLFormer with the LG lifting scheme are more sparse than those with the local lifting scheme. For example, the recognition accuracies (such as church, commercial, parking, square, stadium, storage tanks, and viaduct) are increased while considering the LG relations. By contrast, the effectiveness of the WHU-RS19 is more evident. In a word, it suggests that the LG lifting scheme (parallel) shows better performance than the local one.

As for the convergence, the accuracy and loss curves of the proposed LGLFormer are displayed. As shown in Fig. 10, Fig. 10(a) presents the compare loss curves of LGLFormer on AID validation dataset, Fig. 10(b) gives the compare loss accuracy of LGLFormer on AID validation dataset with 50% data ratio for training. It gives four modes of the LGLFormer, including the local, global, LG (series), and LG (parallel) lifting scheme. It shows that the LGLFormer with global modeling introduction is of better accuracy with faster convergence speed. In addition, the LG (parallel) mode performs better than the series mode in accuracy and convergence on the AID dataset.

E. Complexity Analysis

To compare the advantages of the proposed global and LG lifting schemes, the parameters and the GFlops are figured out here. The proposed methods with local, global, LG (series), and LG (parallel) lifting schemes are listed. In detail, their parameters and the GFlops are shown in Table V.

It shows that the proposed methods, which take global modeling into consideration, only result in a slight increase in parameters and GFlops. In detail, the LG lifting scheme in parallel mode shows an increase of only 0.041M parameters and 0.514 GFlops. Furthermore, the parameters and GFlops of the parallel LG mode are fewer than those of the series LG mode. This is mainly because the convolution of local lifting increases the channels to further calculate global modeling. When local and global lifting processes are in parallel mode, it only considers the original input's channels. Thus, it does not significantly increase the parameters and GFlops required for training.

F. Feature Visualization

The feature visualization results of three scenes are presented in Fig. 11. Four modes are compared, which include the local lifting scheme, the global lifting scheme, and the LG lifting schemes in series and parallel. Specifically, Fig. 11(a)–(c) depicts the feature visualizations of airplanes, rivers, and storage tanks, respectively.

As shown, the global lifting scheme demonstrates a better ability to focus on the object compared to the local lifting scheme. The LG lifting scheme integrates the advantages of both local and global lifting modes. Moreover, the parallel LG lifting scheme excels in focusing on the object while simultaneously considering relevant background information. It effectively retains essential features and reduces redundant features for improved recognition.

V. CONCLUDING REMARKS

In this article, we propose a novel approach called the LGLFormer network, referred to as LGLFormer, specifically designed for remote sensing scene parsing. LGLFormer integrates multiscale and LG representations within the joint framework of CNNs and transformers. Notably, we address the crucial issue of capturing the LG relationships between pixels by introducing a 2-D LGAFE. This novel extractor incorporates global dependencies into the lifting scheme, surpassing the performance of traditional lifting schemes. Furthermore, we propose two distinct LG lifting schemes, including the series mode and the parallel mode.

The LGLFormer has been validated on the benchmark remote sensing datasets. Experimental results demonstrate that the LGLFormer achieves outstanding performance in remote sensing scene classification. In detail, it achieves the state-of-the-art with overall accuracies of 99.02%, 99.2%, and 99.48% on the AID, WHU-RS19, and UCM datasets, respectively. In addition, the ablation experimental results demonstrate the superiority of the LG lifting scheme. Furthermore, the experimental results suggest that the proposed methods possess better convergence ability.

Furthermore, there are still some limitations to LGLFormer. For example, we have only validated its performance on three benchmark remote sensing scene datasets. It is possible to extend its application to other datasets, such as natural images

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and medical images. Moreover, the proposed LGLFormer and the LG lifting scheme exhibit strong representation capabilities. Therefore, they could potentially be extended to other visual tasks, including image segmentation, object detection, and so on.

Hence, future research efforts will focus on exploring the applicability of the proposed method in various feature representation tasks. Moreover, considering the strong representation ability of the LG lifting scheme, integrating it into other deep learning frameworks such as transformer and MLP could be a promising direction for our further study.

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